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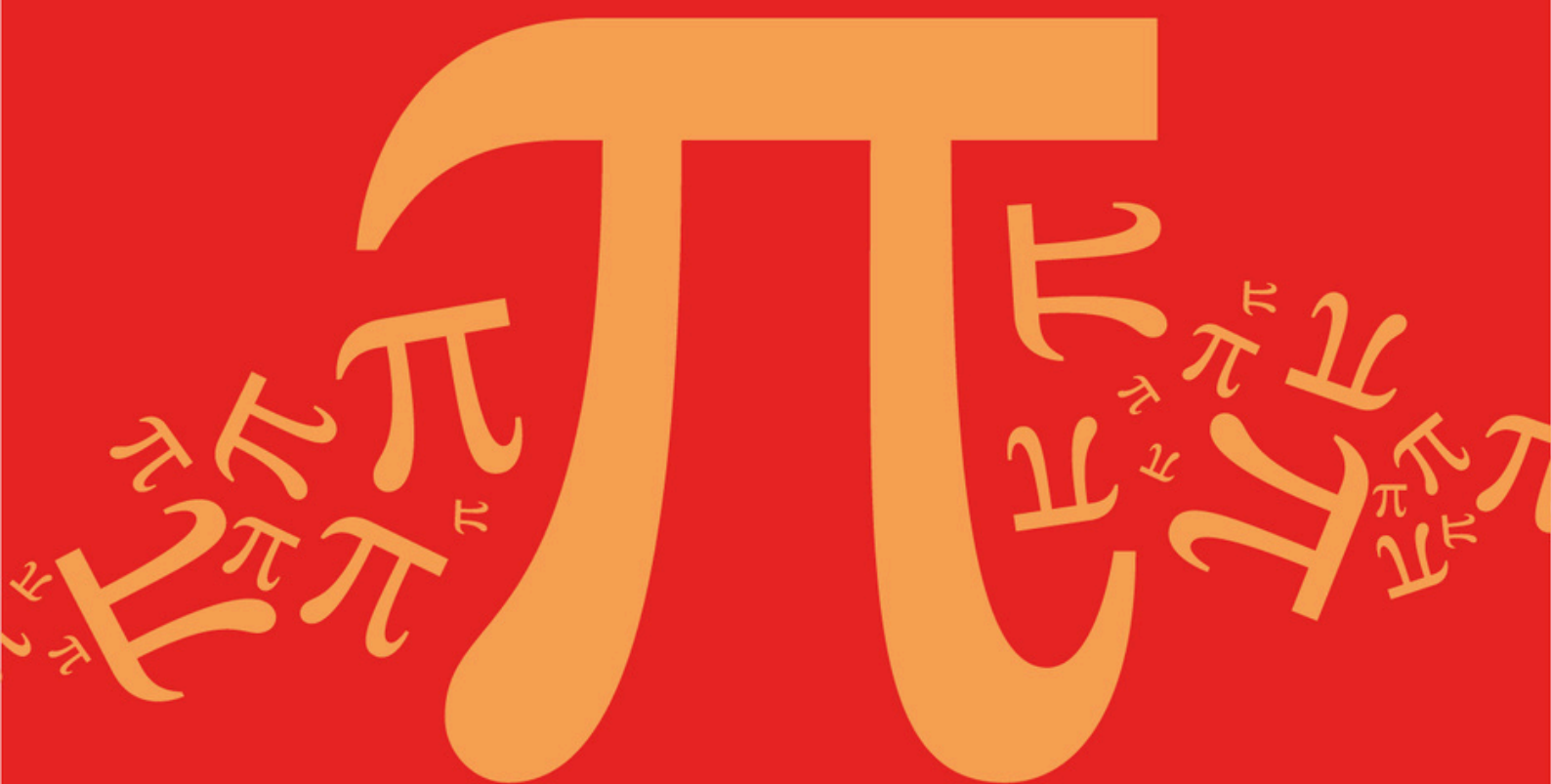
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AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB1: Terminology for Networks

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

DB: Networks

Sub topic

DB1: Terminology for Networks

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

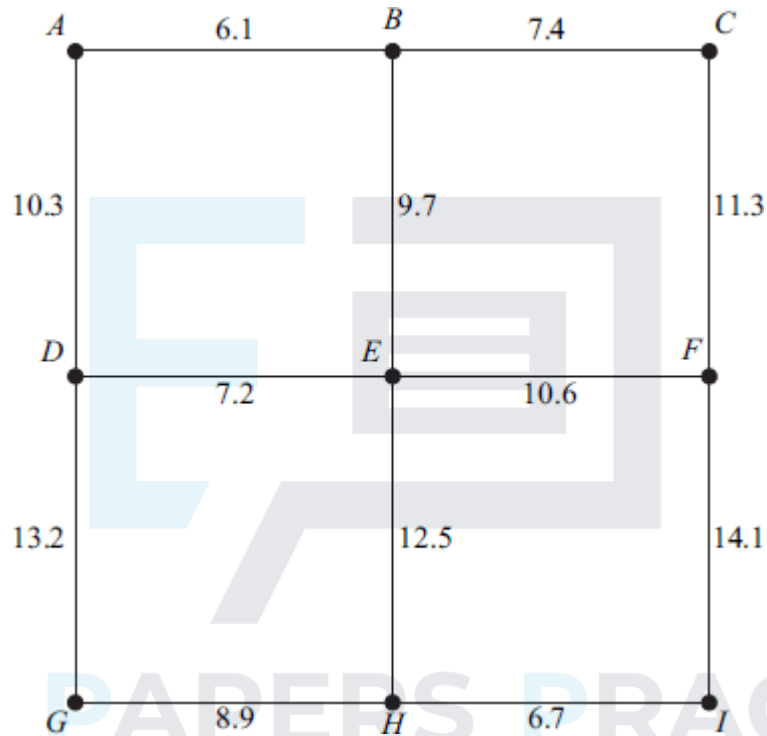
Total Marks Available : _____

Total Marks Received : _____

Q1.

The following network shows the lengths, in miles, of roads connecting nine villages, $A, B, \dots, I$.

A delivery man lives in village A and is to drive along all the roads at least once before returning to A .



Total length of all the roads is 118 miles.

- (a) Find the length of an optimal Chinese postman route around the nine villages, starting and finishing at A .
- (b) For an optimal Chinese postman route corresponding to your answer in part (a), state:
 - (i) the number of times village E would be visited;
 - (ii) the number of times village I would be visited.

(5)

(2)

(Total 7 marks)

Q2.

Tony delivers paper to five offices, A, B, C, D and E . Tony starts his deliveries at office E and travels to each of the other offices once, before returning to office E . Tony wishes to keep his travelling time to a minimum.

The table shows the travelling times, in minutes, between the offices.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>A</i>	–	10	16	20	8
<i>B</i>	10	–	21	15	9
<i>C</i>	16	21	–	10	23
<i>D</i>	20	15	10	–	17
<i>E</i>	8	9	23	17	–

- (a) Find the travelling time of the tour *ACDBEA*. (1)
- (b) Hence write down a tour, starting at *E*, which has the same total travelling time as your answer to part (a). (1)
- (c) Use the nearest neighbour algorithm, starting at *E*, to find an upper bound for the minimum travelling time for Tony's tour. (4)
- (d) By deleting *E*, find a lower bound for the minimum travelling time for Tony's tour. (4)
- (e) Sketch a network showing the edges that give the lower bound in part (d), and comment on its significance. (2)
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Q3.

Sarah is a mobile hairdresser based at *A*. Her day's appointments are at five places: *B*, *C*, *D*, *E* and *F*. She can arrange the appointments in any order. She intends to travel from one place to the next until she has visited all of the places, starting and finishing at *A*. The following table shows the times, in minutes, that it takes to travel between the six places.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	–	15	11	14	27	12
<i>B</i>	15	–	13	19	24	15
<i>C</i>	11	13	–	10	19	12
<i>D</i>	14	19	10	–	26	15
<i>E</i>	27	24	19	26	–	27

<i>F</i>	12	15	12	15	27	–
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- (a) Sarah decides to visit the places in the order *ABCDEF**A*. Find the travelling time of this tour. (1)
- (b) Explain why this answer can be considered as being an upper bound for the minimum travelling time of Sarah's tour. (2)
- (c) Use the nearest neighbour algorithm, starting from *A*, to find another upper bound for the minimum travelling time of Sarah's tour. (4)
- (d) By deleting *A*, find a lower bound for the minimum travelling time of Sarah's tour. (4)
- (e) Sarah thinks that she can reduce her travelling time to 75 minutes. Explain why she is wrong. (1)
- (Total 12 marks)**

Q4.

The network shows the times, in minutes, taken by police cars to drive along roads connecting 12 places, *A*, *B*, ..., *L*.



Each day, a police car has to drive along each road at least once, starting and finishing at A.

(a) find the driving time for the police car;

(6)

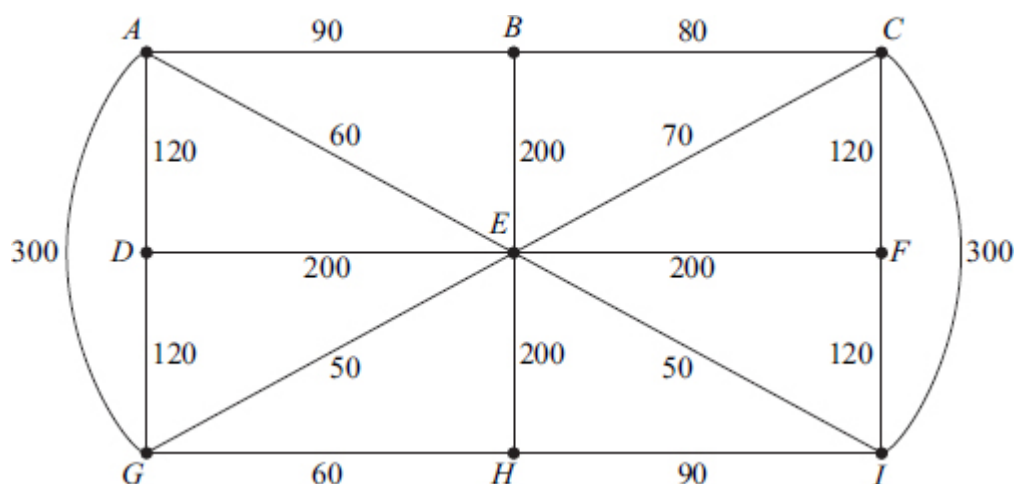
(1)

(Total 7 marks)

The network below shows some streets in a town. The number on each edge shows the length of that street, in metres.

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before returning to his restaurant.



The total length of the streets is 2430 metres.

- (a) Find the length of an optimal Chinese postman route for Tony. (5)
- (b) Colin also wishes to distribute some leaflets. He starts from his house at H , walks along all the streets at least once, before finishing at the restaurant at B .
Colin wishes to walk the minimum distance. Find the length of an optimal route for Colin. (1)
- (c) David also walks along all the streets at least once. He can start at any vertex and finish at any vertex. David also wishes to walk the minimum distance.
 - (i) Find the length of an optimal route for David. (1)
 - (ii) State the vertices from which David could start in order to achieve this optimal route. (1)

(Total 8 marks)

Q6.

A connected graph G has five vertices and has eight edges with lengths 8, 10, 10, 11, 13, 17, 17 and 18.

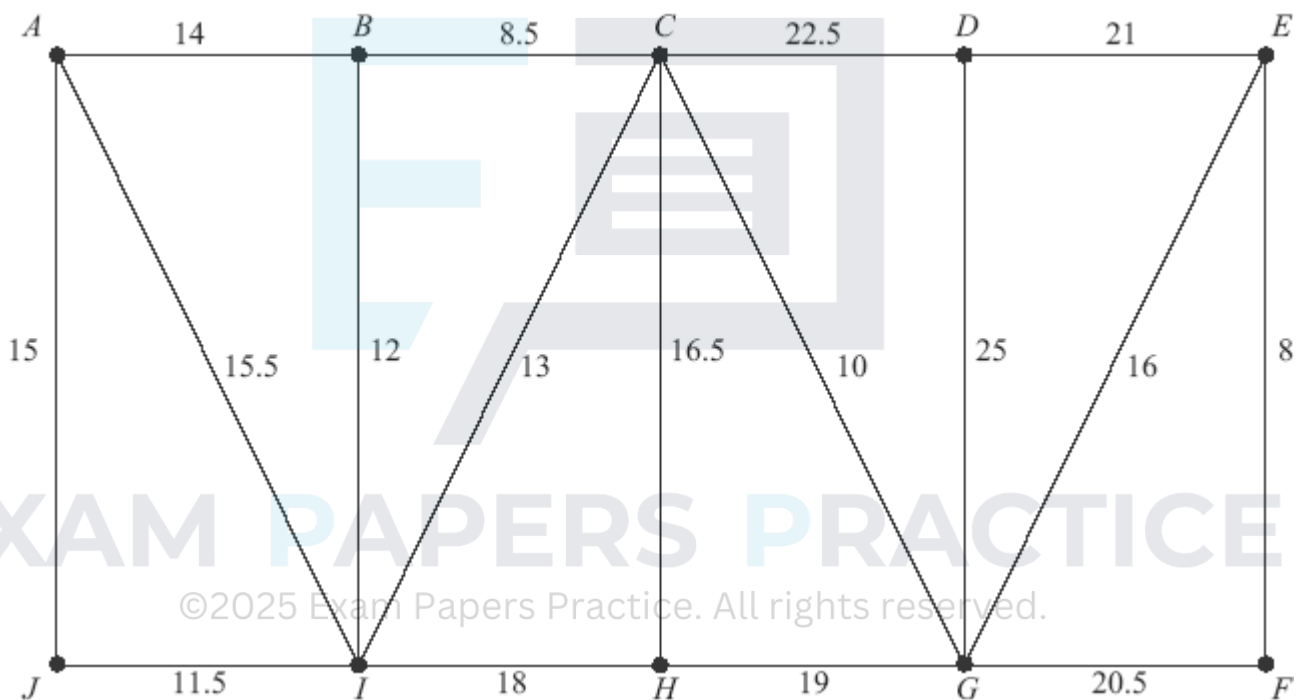
- (a) Find the minimum length of a minimum spanning tree for G . (2)
- (b) Find the maximum length of a minimum spanning tree for G . (2)
- (c) Draw a sketch to show a possible graph G when the length of the minimum spanning tree is 53. (3)

(Total 7 marks)

Q7.

- (a) (i) State the number of edges in a minimum spanning tree for a network with 10 vertices. (1)
- (ii) State the number of edges in a minimum spanning tree for a network with n vertices. (1)

- (b) The following network has 10 vertices: $A, B, \dots, J$. The number on each edge represents the distance between a pair of adjacent vertices.

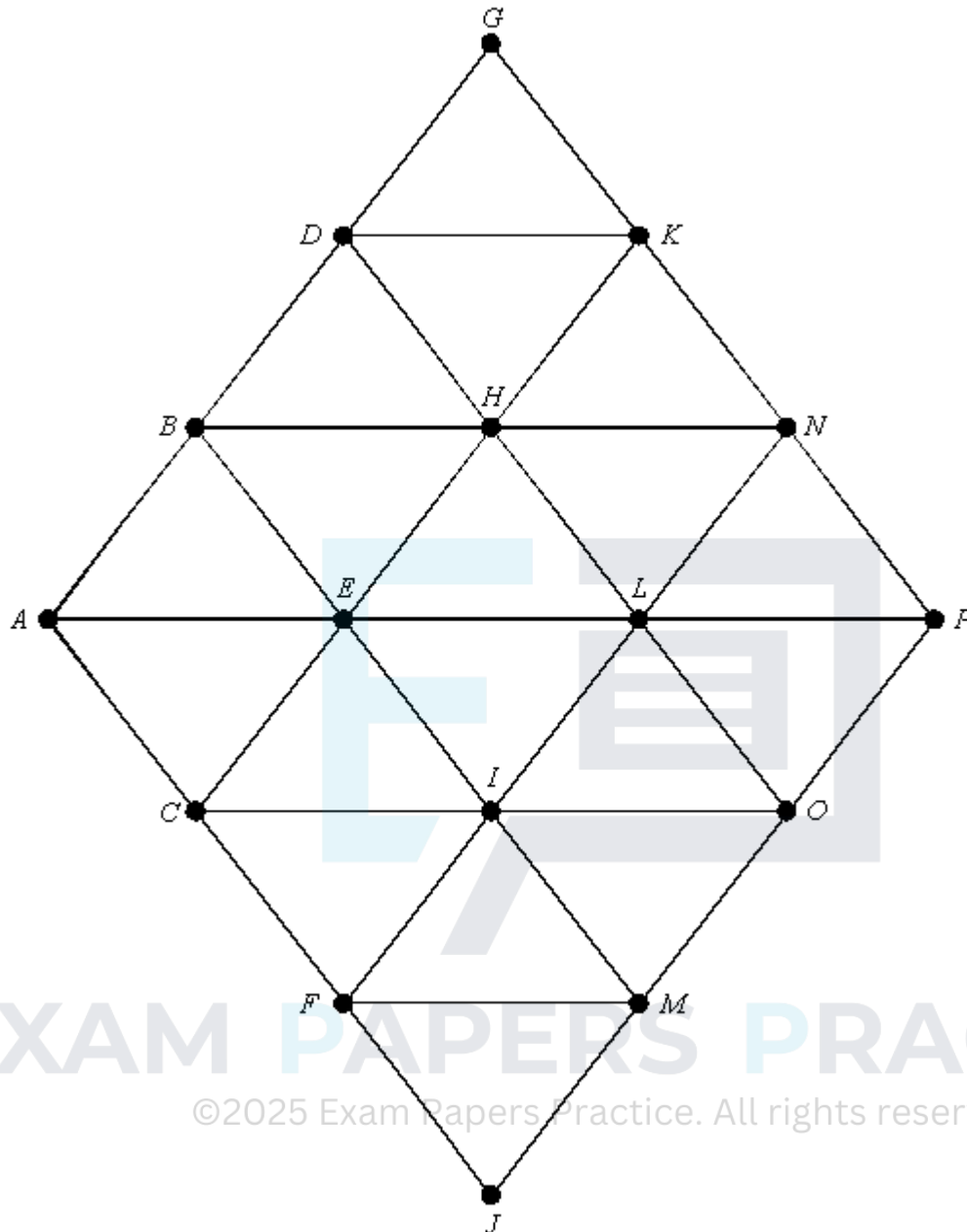


- (i) Use Kruskal's algorithm to find the minimum spanning tree for the network. (5)
- (ii) State the length of your minimum spanning tree. (1)
- (iii) Draw your minimum spanning tree. (2)

(Total 10 marks)

Q8.

The following network has 16 vertices.



- (a) Given that the length of each edge is 1 unit, find:
- (i) the shortest distance from A to K ; (1)
 - (ii) the length of a minimum spanning tree. (2)
- (b) (i) Find the length of an optimal Chinese Postman route, starting and finishing at A . (4)
- (ii) For such a route, state the edges that would have to be used twice. (1)
- (iii) Given that the edges AE and LP are now removed, find the new length of an optimal Chinese Postman route, starting and finishing at A .

(2)
(Total 10 marks)

Q9.

It is the year 2302 and space travel between the planets of the solar system has become commonplace.

Sarah, who is based on Mars, is planning a tour of four other planets before returning to Mars. The planets orbit the sun, and the distance between each pair of planets varies. The following table gives the distance between planets, in space units, in terms of a variable x .

	Mars (M)	Jupiter (J)	Saturn (S)	Neptune (N)	Pluto (P)
Mars (M)	-	$3x + 6$	$x + 24$	$2x + 17$	$5x$
Jupiter (J)	$3x + 6$	-	$x + 4$	$2x - 3$	$3x - 2$
Saturn (S)	$x + 24$	$x + 4$	-	$x + 13$	$2x - 1$
Neptune (N)	$2x + 17$	$2x - 3$	$x + 13$	-	$x + 6$
Pluto (P)	$5x$	$3x - 2$	$2x - 1$	$x + 6$	-

- (a) Given that Jupiter is the nearest planet to Mars, find three inequalities in x .

(3)

Sarah plans a tour using the nearest neighbour algorithm and finds that, for one particular **integer** value of x , this algorithm gives a **unique** tour of $M J S P N M$.

- (b) By considering the fact that Sarah's tour visits Saturn immediately after Jupiter, find two further inequalities in x .

(3)

- (c) Hence find the value of the integer x , and the total length of Sarah's tour.

(5)

(Total 11 marks)

Q10.

- (a) A connected graph has four vertices. State the number of edges in the graph's minimum spanning tree

(1)

- (b) A graph has n vertices. The graph is complete, i.e. each vertex is joined to every other vertex by exactly one edge.

- (i) State the number of edges in the graph's minimum spanning tree

(1)

- (ii) Determine the number of Hamiltonian cycles in the graph.

(2)

- (c) A connected graph has four vertices and has arc lengths of
4, 4.5, 5, 6.5, 7, 8 and 9 units.

The length of its minimum spanning tree is 17 units. Draw a sketch to show a possible graph.

(3)

(Total 7 marks)

Q11.

- (a) Draw a graph to represent the following matrix.

From \ To	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	0	2	1	0
<i>B</i>	1	0	1	1
<i>C</i>	1	1	2	2
<i>D</i>	1	0	3	1

(3)

- (b) Explain from the matrix why your graph must be a directed graph.

(1)

(Total 4 marks)